

Evaluating LLMs

Generative AI

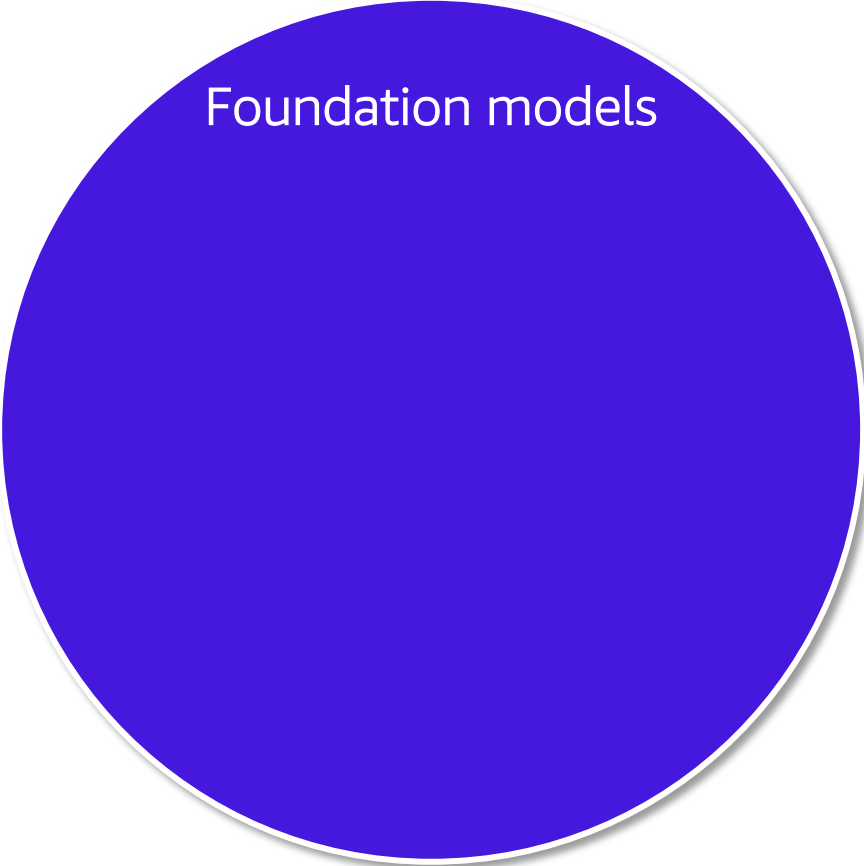
Module 2 – Lesson 1

Today's activities



- Review of module 1
- Why is it important to evaluate LLMs?
- Evaluating LLMs
- Performance Evaluation on Amazon Bedrock

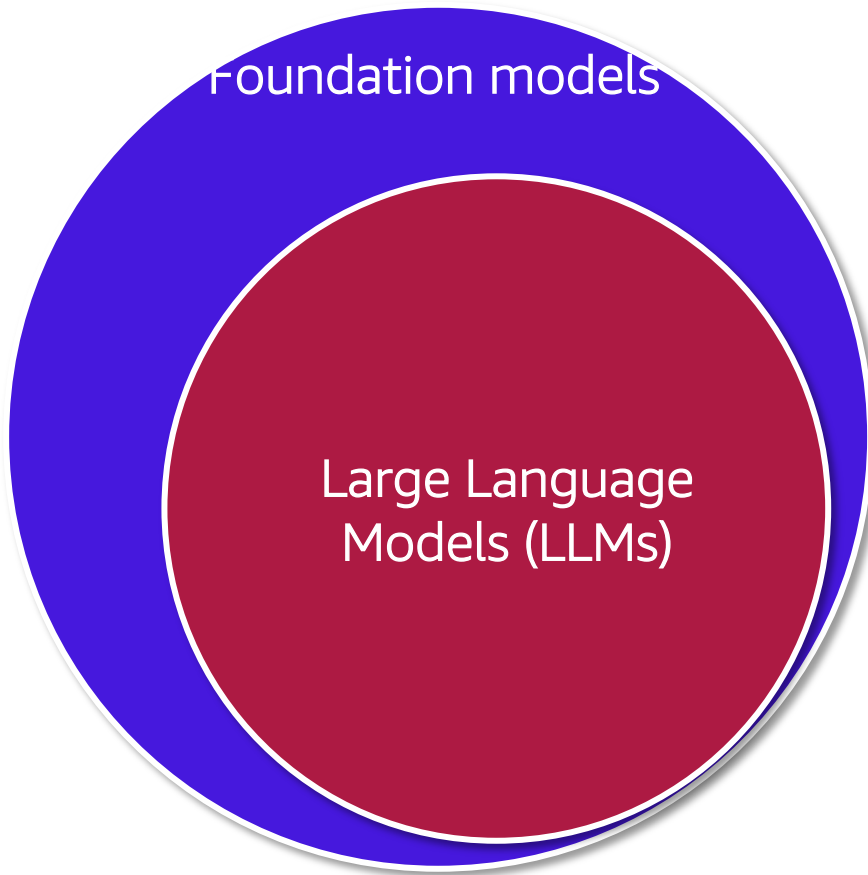
Review: Foundation models



Foundation models

- Large ML models that are **pre-trained** with **vast amounts of data**. These can be **adapted** to more specialized tasks.
- Can be trained on any kind of data
 - Text
 - Images
 - Video
 - Audio
 - Etc...

Review: Large Language Models (LLMs)



- Foundation models **trained on text**.
- Large ML models that learn the **probabilities of words** being used in certain contexts.
- **Training task:** Learn to predict the missing word in a text sequence.

"The weather has been cloudy for the last two days. Most likely it will be ____ tomorrow."

cloudy? sunny? foggy?

Review: Good prompting practices

- Write **clear** and **specific** instructions (unambiguous and specific).
- Highlight or specify the part of the prompt **that the model should focus** on.
- Add relevant **details** or **restrictions** to your prompt.
- Separate the instruction, content, question, and output directions.
- Prefer using positive instructions.
- Finding the optimum prompt is usually an **iterative** process which may take a few attempts.

Why is it important to evaluate LLMs?

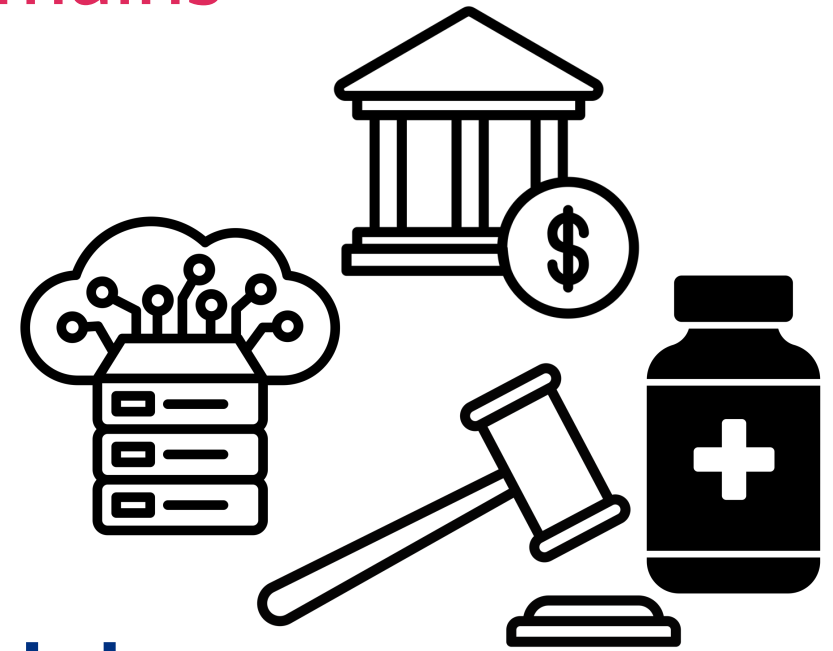
1. Increasing number of core abilities

- LLMs have **increasing number** of core abilities:
 - Question-answering
 - Content generation
 - Logical reasoning
 - Arithmetic reasoning
 - Common sense reasoning
 - Code generation
 - Multi-hop reasoning
 - and more!



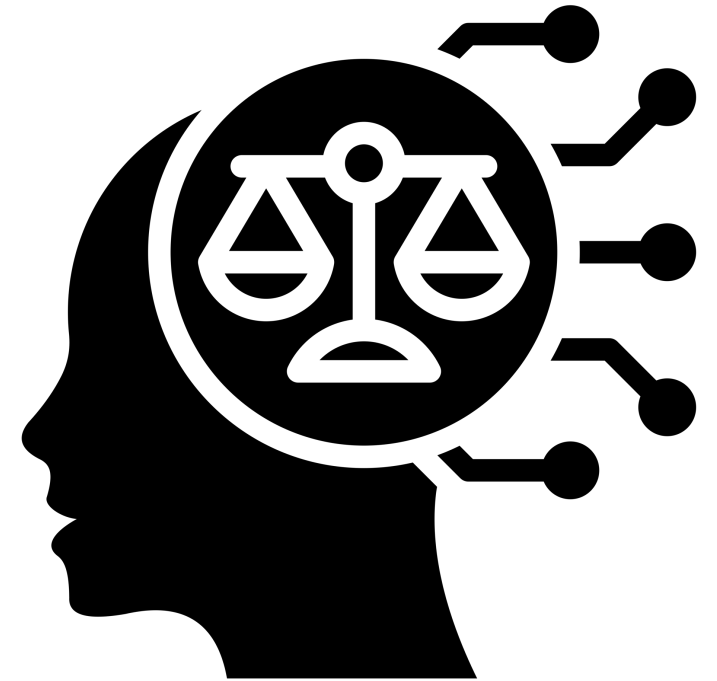
2. Knowledge of diverse topics

- LLMs have knowledge **about various domains**
 - LLMs are trained on diverse data
 - Finance
 - Education
 - Software development
 - Medicine
 - etc...
 - However, LLMs can have **shallow breadth of knowledge**
 - Does their performance align with the company's data, policies and brand?



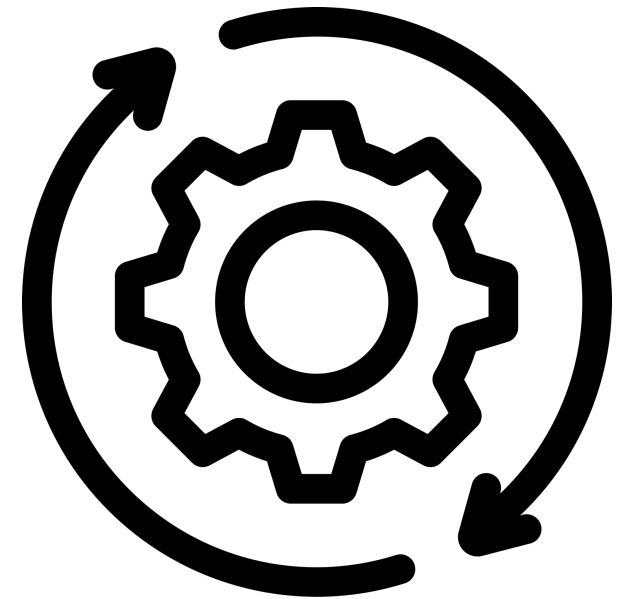
3. Increasing visibility and influence

- **Increasing influence** of LLMs on societal impact
 - Increasingly used to interact with information
 - Safety considerations
 - Security concerns
 - Ethical implications
- **Responsible AI is key!**



4. Going beyond content generation

- Agentic applications empower **LLMs to execute actions**
 - Write and execute code
 - **Security policies need to be followed!**
 - Natural language-to-query generators for databases
 - Invoke API calls
 - Robotic control and manipulation
 - System access
 - **Agent-based Operating System (OS)**



5. LLMs Acting as experts for other systems

- **Evaluators** for other systems
- Used to **annotate and label data**
- Used to **evaluate other LLMs**



Evaluating LLMs

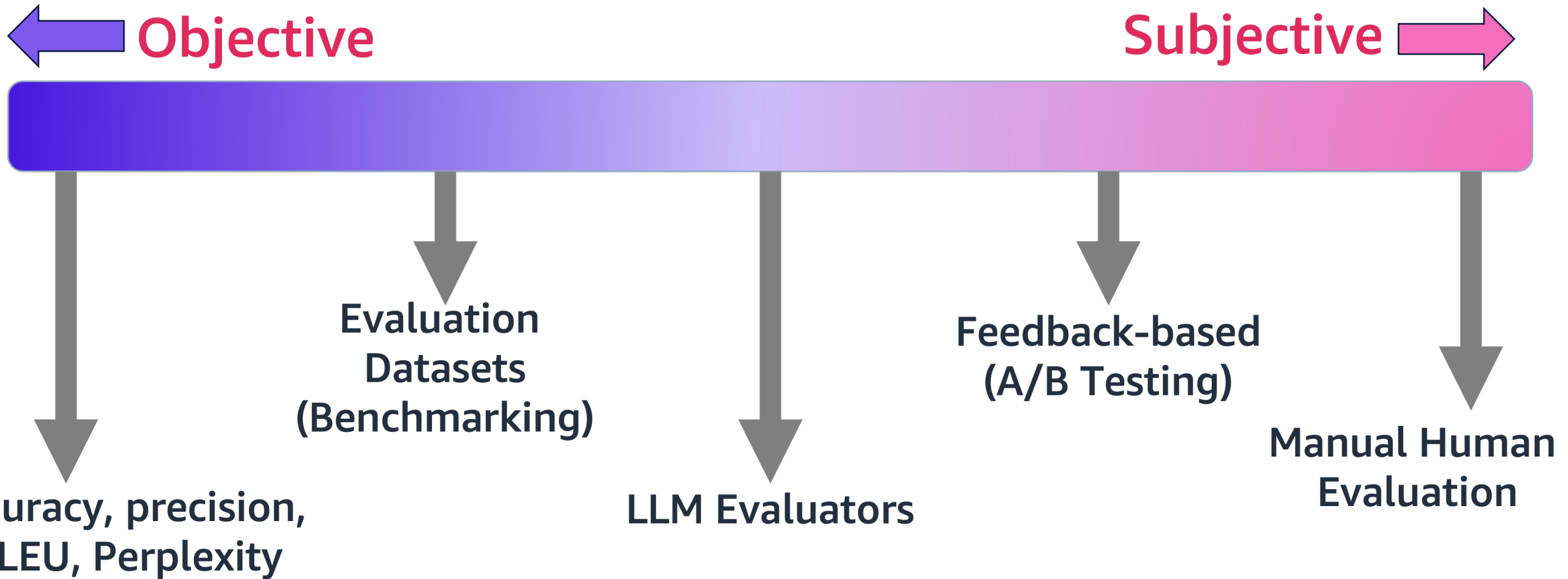
Evaluating LLMs is hard

- Evaluating language is quite **subjective**
- LLMs are extremely **versatile** tool
 - Comprehensive evaluation is difficult
- No single **universal evaluating framework** or protocol
- Need high quality data for evaluation
- Computational constraints to evaluate these models

Where do LLMs fail?

- Complex arithmetic problems and analytical reasoning
- Reliable knowledge source for task specific topics
- Implicit understanding and mitigation of biases
- Potential for generating false, misleading or manipulative content
 - Hallucinations
- Analyzing structured data
 - Databases, spreadsheets, etc.
- Compliance with every rule, guidelines and protocol

Spectrum of evaluation approaches



Metric-based evaluation

- **Objective metrics** to evaluate LLM performance
- Do not extend easily to evaluate complex traits such as reasoning
- A few examples:
 - **Accuracy**: Proportion of correct prediction
 - **Perplexity**: Likelihood of predicting the given text next
 - **BLEU**: Precision-based text comparison
 - **ROUGE**: Recall-oriented text evaluation

Evaluation Datasets

- Use of high quality benchmarking datasets
- Test more **complex traits** leveraging objective metrics
 - Common sense reasoning
 - Arithmetic reasoning
 - General knowledge
 - Software development
 - Bias
 - Toxicity...

Examples of evaluation datasets

- **Massive Multitask Language Understanding (MMLU)**
 - Measures general knowledge across 57 different subjects, ranging from STEM to social sciences
- **HellaSwag**
 - Tests natural language inference in a way that requires understanding of intricate details
- **GSM8K**
 - A set of 8.5K grade-school math problems that require basic to intermediate math operations
- **AGIEval**
 - A collection of standardized tests, including GRE, GMAT, SAT, LSAT, and civil service exams
- **Responsible AI (RAI)**
 - A framework for evaluating the safety of chat-optimized models in conversational settings

HellaSwag: Example

- Determining who has the right of way:

Come to a complete halt at a stop sign or red light. At a stop sign, come to a complete halt for about 2 seconds or until vehicles that arrived before you clear the intersection. If you are stopped at a red light, proceed when the light has turned green...

A: Stop for no more than 2 seconds, or until the light turns yellow. A red light in front of you indicates that you should stop



B: After you come to a complete stop, turn off your turn signal. Allow vehicles to move in different directions before moving onto the sidewalk



C: Stay out of oncoming traffic. People coming in from behind may elect to stay left or right



D: If the intersection has a white strip in your lane, stop before crossing this line. Wait until all traffic has cleared before crossing the intersection

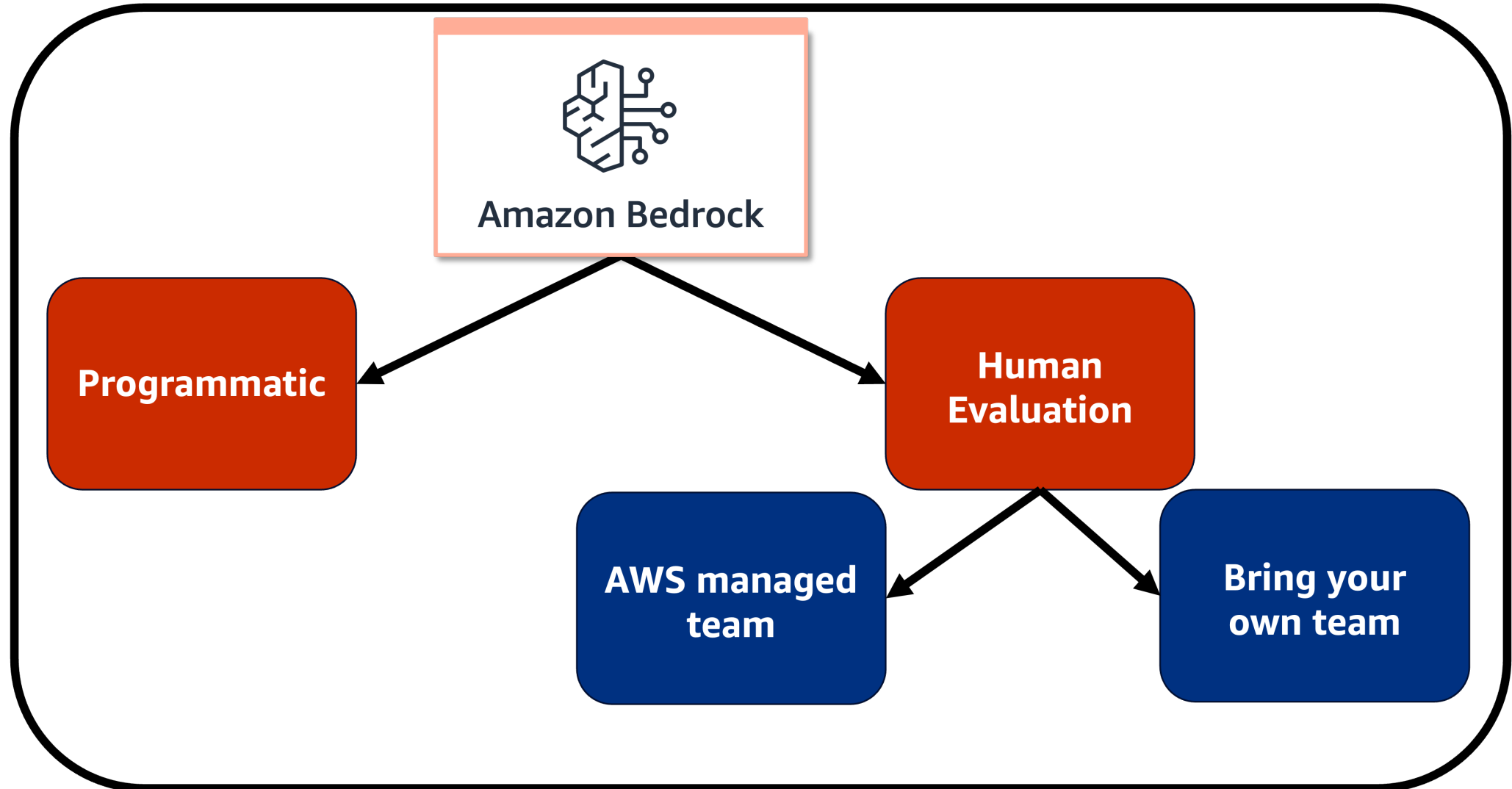


Always evaluate, even if difficult

- **Never use an AI-generated solution without having evaluated it**
- Evaluating LLM outputs is hard
 - Focus on the metrics and benchmarks that are important for your use case
 - You will probably need to collect your own annotated benchmark dataset
 - In certain cases, automated metrics (accuracy, perplexity, ...) might be enough to assess the LLM performance
 - It is likely you will need to engage human evaluators: you can use Amazon Bedrock's Human-In-Loop (HIL) capabilities

Performance Evaluation on Amazon Bedrock

Evaluation on Amazon Bedrock



Programmatic evaluation



Amazon Bedrock



Choose a FM



Select task
type



Choose metric



Select dataset
or upload a
dataset

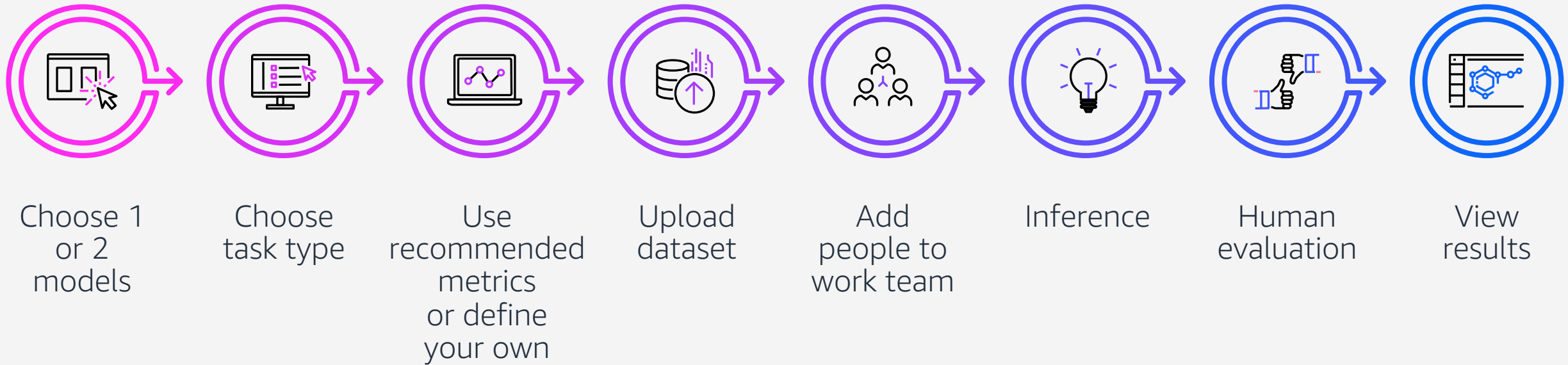
1. Select a foundation model from Amazon Bedrock
2. Select the type of task for evaluation
 - Text generation, classification, Q&A, etc.
3. Choose the set of metrics
 - Accuracy, Robustness, Toxicity, etc.
4. Select an available dataset or upload your own prompt dataset

Manual evaluation

- Human evaluations supported by Amazon Bedrock
- Evaluate responses from up to two models
- Two options for evaluation:
 - Bring your own team
 - AWS Managed work team

How human evaluation works

(Bring your own team)



Analyze results

Accuracy

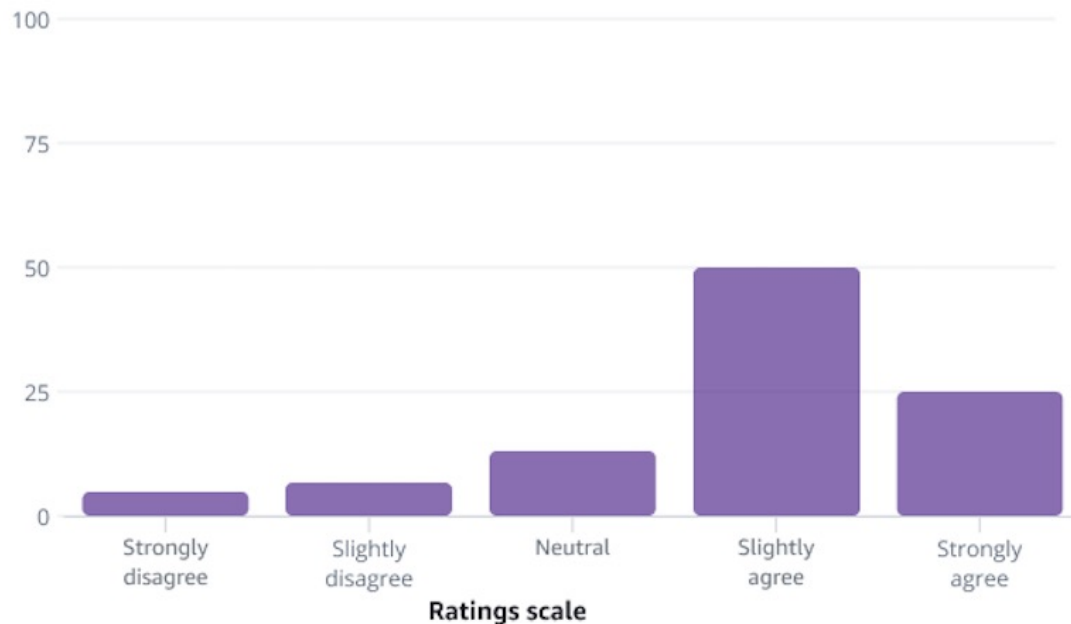
Evaluation type
Individual

Rating mechanism
Likert scale - Individual

Metric type:

Approval strength. A 5 point Likert scale of how strongly the evaluator agreed that the response from the model met the evaluation criteria.
1 strongly disagree, 5 is strongly agree.

Model A
Number of ratings



Model B
Number of ratings



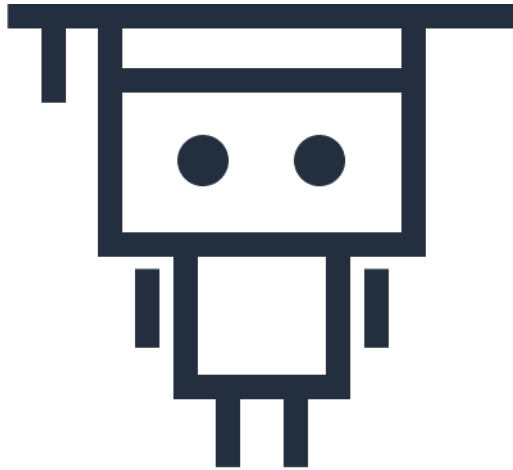
Evaluation report:

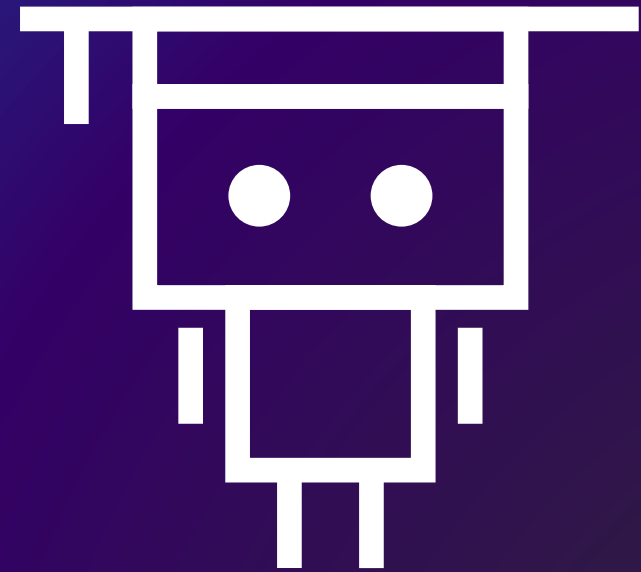
- ✓ Track ratings from your team
- ✓ See distributions visually
- ✓ Simple metric explanations



Next lesson

- This lesson covered topics related to evaluation of LLMs
- In the next lesson, you will learn about foundations of responsible AI.





Thank you!